Positron spectra

Integrating Eq. (4) over a spherically symmetric SN model, we obtain the positron spectrum as

$$\frac{dN_{\rm pos}}{dE_{\rm pos}} = \sum_i n_i B_i \left[\int dt \int_0^\infty 4\pi r^2 dr \, \frac{d^2 n_a}{dE_a \, dt} \times \left(\epsilon_{II} \, e^{-r_{II}/\lambda_i(E_a,r)} + \epsilon_I \, e^{-r_I/\lambda_i(E_a,r)} \right) \right] \Bigg|_{E_a = x_i E_{\rm pos}} , \tag{21}$$

where r is the radial coordinate from the center of the SN core, t is integrated over the FIP emission time, i.e. $1-10$ s, $\lambda_i(E_a, r)$ is the mean-free-path for the i-th decay channel at a given FIP energy and location, which produces n_i positrons with a branching ratio B_i and average energy E_a/x_i. In the electrophilic axion case, only $a \to e^+e^-$ decays are allowed, therefore $n_1 = 2$, $B_1 = 1$ and $x_i = 2$. Moreover, following Ref. [86] we fix

$$r_{II} = 10^{14} \text{ cm}, \qquad r_I = 2 \times 10^{12} \text{ cm}, \tag{22}$$

for the envelope radii of Type II and Ib/c SNe, while according to Ref. [89], we take as average fractions of SNe of Type II and Ib/c

$$\epsilon_{II} = 1 - \epsilon_I, \qquad \epsilon_I = 0.33 \ . \tag{23}$$

III. SECONDARY γ-RAY PRODUCTION AND COMPARISON WITH MEV DATA

Injection and propagation of $e^\pm$ from FIPs

The decay into electrons and positrons from any FIP produced in SNe leads to a continuous diffuse sea of e^+e^- pairs with energies of tens to hundreds of MeV, which are confined in the Galaxy for Myrs [25, 26]. Given that the rate of SN explosion is negligible compared to the propagation time of these particles, their injection can be considered smooth and continuous and following the spatial distribution of SN remnants. Once the e^+e^- are injected into the ISM, these particles interact with the Galactic environment, being scattered by magnetohydrodynamic fluctuations in the ISM plasma [90–93], which makes them propagate diffusively in the Galaxy. During their propagation, their interaction with the Galactic magnetic field, gas and radiation fields produces secondary radiations and makes these particles lose energy.

Following a similar procedure to Refs. [26, 94], we use a customized version [95] of the `DRAGON2` code [96, 97], a dedicated CR propagation code prepared to simulate CR diffusion, accounting for all diffusion-reacceleration-advection-loss effects in the propagation of Galactic CRs [90]. We simulate electron-positron signals for different kinds of FIPs produced in SNe in the range of kinetic energies from 100 eV to 5 GeV, with an energy resolution of 5%.

We compute the steady-state diffuse distribution (in space and energy) of the electrons and positrons produced from the different FIP models studied here by solving their propagation equation numerically. In particular, we follow the same procedure illustrated in Refs. [26, 94, 100], where we refer the reader for more details. The steady-state solution for the distribution of these particles in the Galaxy can be obtained by solving the differential diffusion-reacceleration equation imposing time-independent $e^\pm$ density in each point of the Galaxy ($\frac{dn_e}{dt} = 0$)

$$-\nabla \cdot \left(D \vec{\nabla} f_e \right) - \frac{\partial}{\partial p_e} \left[\dot{p}_e f_e - p_e^2 D_{pp} \frac{\partial}{\partial p_e} \left(\frac{f_e}{p_e^2} \right) \right] = Q_e \ . \tag{24}$$

In this equation, p_e is the momentum and therefore $f_e \equiv \frac{dn_e}{dp_e}$ is the density of $e^\pm$ per unit momentum at a given position. Spatial diffusion is characterized by the diffusion coefficient D, which basically follows a power-law in rigidity. This diffusion coefficient can be precisely determined at GeV energies for analyses of CR nuclei [101–103], however, uncertainties can be important at lower energies. We adopt a set of propagation parameters that allow us to reproduce all current CR observables at the GeV scale, adopting the spiral arm distribution of the gas. These are the propagation parameters used in Ref. [94], which reports detailed discussions on this choice. We also examine the effect of uncertainties in these parameters below (see Fig. 7 and the discussion around it). In Eq. (24) we take into account all the sources of momentum losses $\dot{p}_e$ due to interactions with the Galactic environment, which are mainly synchrotron interactions with the Galactic magnetic field, inverse Compton scattering off CMB light and Galactic radiation fields, ionization of molecular gas, and Coulomb interactions and bremsstrahlung with the ISM plasma. Momentum diffusion (or reacceleration) is also included through the term $D_{pp} \propto \frac{v_A^2 \cdot p^2}{D}$. This term becomes the main source of uncertainty in the predicted $e^\pm$ spectrum for particles injected with energies below a few tens of MeV, as shown in Refs. [25, 26, 94]

Finally, the source term, Q_e, regulates the injection of $e^\pm$. It is described with a spatially dependent term featuring the spatial distribution of SNe, and the energy-dependent injection term, which is calculated imposing that the injected number of particles per unit energy is equal to the integral of the total flux density of particles over the volume of the Galaxy, as described in Ref. [26]. In this calculation, we assume that SNe follows the Ferriere distribution [104], convolved with the Steiman-Cameron distribution [105] of the spiral arms (four-arm model). We have checked that the effect of using a different spatial distribution (concretely, the Lorimer distribution [106]) leads to uncertainties always below 20% in the predicted $e^\pm$ flux. We have computed tables with the $e^\pm$ injection spectra from each class of FIP, which are used as input